

UTSAV BK

Washington, D.C. | utsavbk9022@gmail.com | [Portfolio](#) | [GitHub](#) | [LeetCode](#)

EDUCATION

Howard University

Bachelor of Science in Computer Science

Aug 2026 – Present

Washington, D.C.

- **Honors & Fellowships:** Howard University Presidential Scholar (Full Merit Scholarship recipient).

Budhanilkantha School

Cambridge International A-Levels

Jun 2023 – Jun 2025

Kathmandu, Nepal

- **Academic Performance:** Physics (A*), Chemistry (A), Mathematics (A), Biology (A), General Paper (a).
- **Honors & Testing:** Morgan Award Winner 2025 (Highest Cambridge A-Level Biology score cohort-wide); SAT Score: 1590/1600 (Math: 790, Reading/Writing: 800).

TECHNICAL PROJECTS & RESEARCH

Founder & Lead Developer — SATitude ([satitude.xyz](#))

Jul 2026 – Present

- Architected and deployed a full-stack, AI-powered digital SAT preparation platform featuring adaptive diagnostic modules and custom performance analytics.
- Engineered an adaptive AI tutor integrating OpenAI REST APIs, Node.js, Express, and PostgreSQL to deliver real-time data streaming and personalized study paths.
- Optimized client-side rendering and SQL query indexing, reducing page load latency by 35% and maximizing search engine discoverability.

Hardware & Software Developer — Hexapod Autonomous Robot

Budhanilkantha Makerspace

- Engineered and assembled a 6-legged autonomous robot under faculty mentorship, integrating microcontrollers and multi-sensor arrays.
- Programmed inverse kinematics and gait-coordination algorithms in Python for smooth multi-axis leg movements and real-time sensor processing.
- Designed custom 3D structural components in Blender and systematically resolved hardware-software timing integration challenges.

Mathematics Research Team Member — Guidelight Nepal

Kathmandu, Nepal

- Selected for an elite 7-person research cohort guided by academic research mentors from Cornell University and the University of Bristol.
- Investigated advanced functional transformations and non-linear parameter behavior using analytical mathematical modeling techniques.

TEACHING & LEADERSHIP EXPERIENCE

Online STEM Tutor — Merosiksha

Aug 2025 – Present

- Delivered comprehensive 1-on-1 and group instruction in Physics, Chemistry, Math, and Biology on Nepal's premier online learning platform.
- Designed customized practice problem sets and step-by-step conceptual breakdowns to improve student mastery of complex STEM material.

Vice President — Environment Club, Budhanilkantha School

Feb 2024 – Feb 2025

- Directed a year-long environmental initiative engaging 1,100+ students through interactive workshops in collaboration with Kathmandu's District Forest Office.

Volunteer Educator — Navigo Nepal ("What's Next Program")

2025

- Educated 500+ high school students across Kathmandu on post-secondary academic pathways, competitive STEM olympiads, and international scholarships.

TECHNICAL SKILLS & SPOKEN LANGUAGES

Languages: Python, C++, JavaScript (ES6+), HTML5, CSS3, SQL

Frameworks & Tools: React, Node.js, Express, REST APIs, PostgreSQL, Git, GitHub, Blender (3D Modeling)

Core Concepts: Data Structures & Algorithms, Asynchronous Data Flow, Kinematics & Robotics Control

Spoken Languages: English (Fluent), Nepali (Native), Hindi (Fluent)